Abstract

This report documents the experimental development of a two-stage computer vision pipeline for localising and recognising pool balls. I first tested a green-channel boost followed by adaptive thresholding and a K-Means colour-clustering classifier; both approaches failed for distinct reasons and were discarded. I then adopted a multi-channel CIELAB edge-fusion detector and a saturation-dominant one-dimensional histogram classifier. The final pipeline achieves an overall Stage 1 precision of 0.886, recall of 0.867, and F1-score of 0.876, with a Stage 2 recognition accuracy of 70.5% across the six test tables.

1 Introduction

The task was to localise every pool ball on a rectified table image and then assign each surviving detection to one of the sixteen known balls. The six test images contain strong shadows, uneven lighting, tightly packed racked balls, and a mixture of solids and stripes. I developed the final pipeline through two explicit experimental failures followed by two redesigns. This report records that journey, including the visual evidence that motivated the final design choices.

2 Experimental Failures

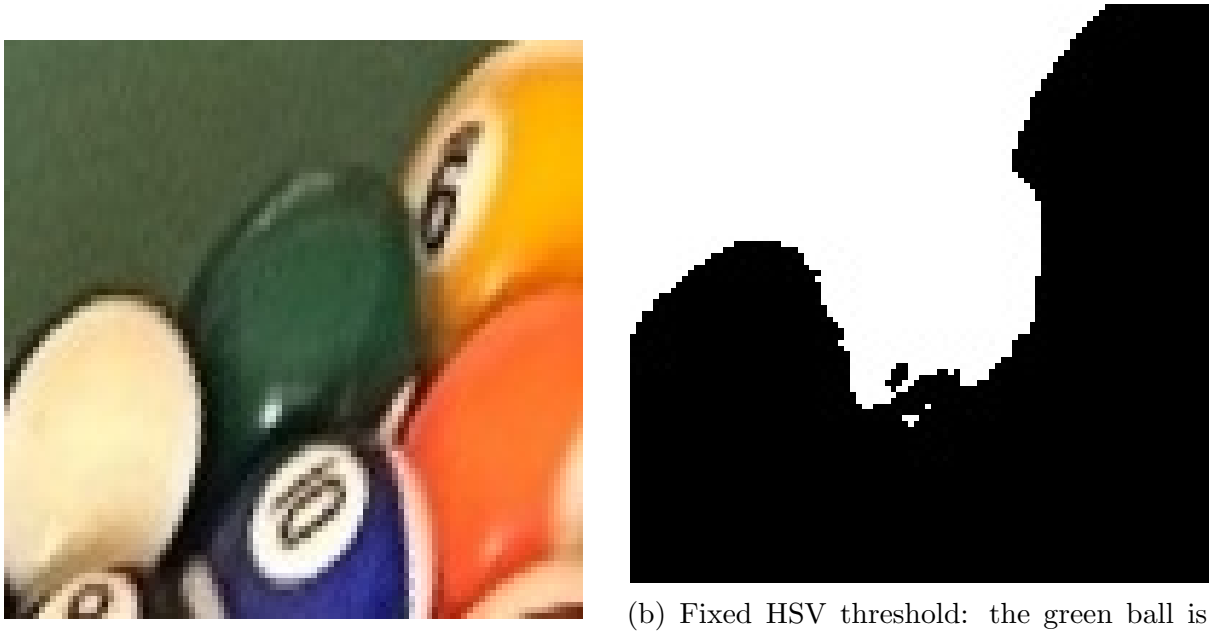
2.1 The Failure of HSV Thresholding

To solve the problem of missing green balls, I initially hypothesised that isolating and boosting the RGB green channel by a factor of 1.5, followed by adaptive Gaussian thresholding, would force the balls to pop against the felt. The adaptive threshold used a local block size of 15 and a constant of 2. This approach failed because boosting the green channel equally boosted the green table, and adaptive thresholding amplified the woven texture of the felt into severe noise. The resulting edge map contained many false circular responses and did not provide a stable basis for detection.

I then attempted to isolate the felt directly in HSV space with a fixed hue range of 35–85, saturation range 40–255, and value range 40–255. This also failed for the green object balls, particularly balls 6 and 14. Their hue and saturation overlap almost completely with the green felt, so the threshold classifies them as table rather than as separate objects. Figure **??** shows a 100×100 region around the green ball on the left and the resulting binary mask on the right. In the mask, the green ball vanishes into the white felt region instead of producing a distinct foreground blob.

2.2 The K-Means Colour-Clustering Attempt

To differentiate striped from solid balls, I first attempted to use K-Means clustering with $K = 2$ on the pixels inside each ball mask to extract the two dominant colours, and then compared those colours with the reference balls using Euclidean distance. This proved computationally heavy and unreliable, because the visibility of the white stripe



(a) Original region around the green ball.

(b) Fixed HSV threshold: the green ball is merged into the felt.

Figure 1: Visual evidence for the failure of fixed HSV thresholding. The green ball and the green felt are not separable in hue–saturation space with static limits.

depends heavily on the physical rotation of the ball. A stripe may be almost hidden at one orientation and dominant at another, so two dominant colour clusters do not form a stable descriptor.

3 Final Pipeline

3.1 Stage 1: Multi-Channel LAB Edge Fusion

I realised that the solution required looking at colour boundaries, not just brightness. I split the rectified image into the CIELAB colour space and applied Gaussian smoothing with a 5×5 kernel and a standard deviation of 1.2 to all three channels. I then ran Canny edge detection independently on each channel. I used thresholds of 45 and 95 for the lightness channel L, and more sensitive thresholds of 35 and 85 for the chromatic channels a and b. The three binary edge maps were combined with a logical OR to produce a highly robust gradient map that captured subtle colour shifts as well as luminance boundaries. This combined map was passed to the Hough Circle Transform with a minimum centre separation of 40 pixels and a ball-radius range of 19 to 24 pixels. Figure **??** shows the combined edge map for Table 4, and Figure **??** shows the corresponding rectified table.

3.2 Geometric Constraints

Two geometric constraints remove candidates that cannot correspond to balls resting on the playing surface. First, a pocket constraint rejects any centre within 95 pixels of the six theoretical pocket positions. Second, a cushion constraint rejects centres within 22 pixels

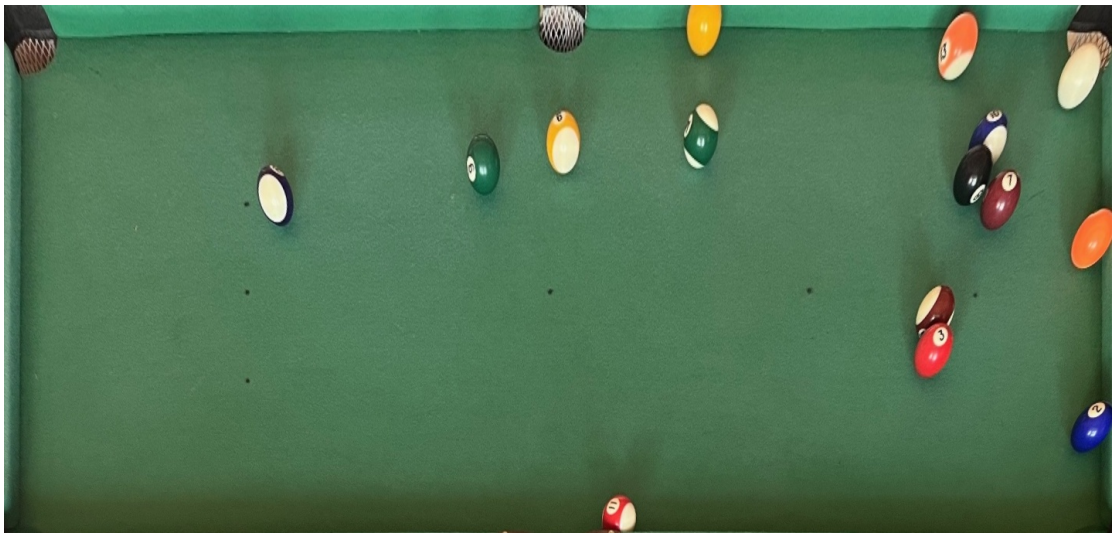


Figure 2: Rectified Table 4 used for the Stage 1 experiments.

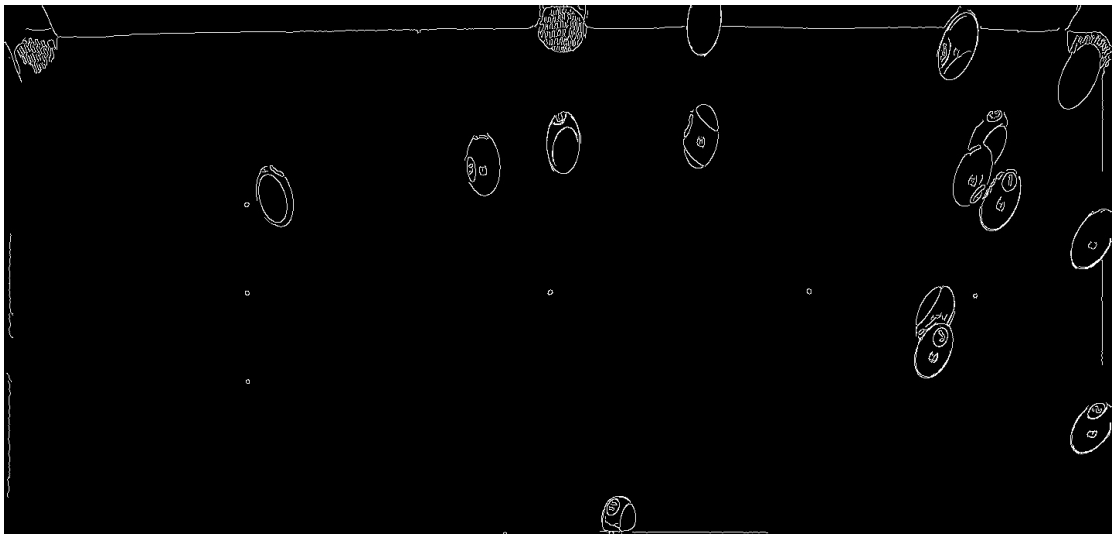


Figure 3: Combined LAB multi-channel edge map for Table 4. The lightness and chromatic channels contribute complementary boundaries, including low-contrast edges of balls hidden in shadow.

of the outer border, because a ball frozen to the cushion has its centre approximately one radius from the cushion and the wooden rails themselves generate non-felt edges. The minimum centre separation of 40 pixels suppresses duplicate detections on the same ball while preserving genuinely adjacent balls in a tightly packed rack.

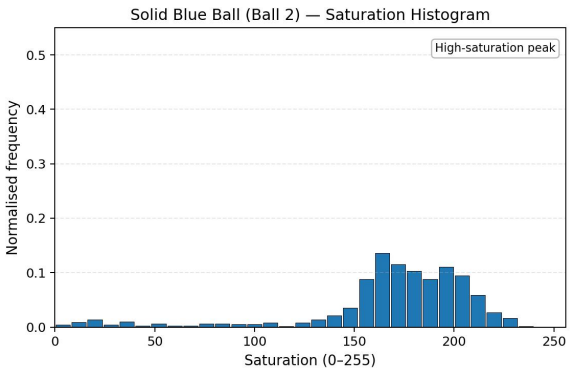
3.3 Stage 2: Saturation-Dominant Histograms

I pivoted from colour clustering to one-dimensional histograms. I extracted a hue histogram with 30 bins and a saturation histogram with 32 bins from a reduced circular sampling mask of radius 16 pixels. The reduced mask prevents a slightly off-centre detection from admitting felt, shadow, or bounding-box corner pixels into the descriptor. I then compared test histograms with the sixteen reference histograms using the Bhattacharyya distance.

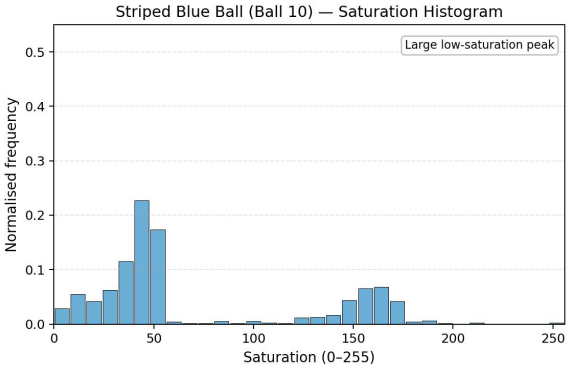
Because a white stripe drastically lowers a ball's average saturation regardless of rotation, I applied a custom weighting system: 30% weight to hue, 70% weight to saturation, and 0% weight to brightness. The final distance is

$$d = 0.30\, d_{\text{hue}} + 0.70\, d_{\text{sat}}. \tag{1}$$

A detection whose weighted distance exceeds 0.80 is rejected as a non-ball or as an unreliable observation. Figure **??** compares the saturation distributions of the solid blue ball 2 and the striped blue ball 10. The solid ball peaks at high saturation, whereas the striped ball has a pronounced low-saturation peak caused by its white stripe. This shift is independent of the ball's physical rotation and provides a mathematically robust feature for distinguishing stripes from solids.



(a) Solid blue ball 2: high-saturation peak.



(b) Striped blue ball 10: large low-saturation peak.

Figure 4: The white stripe substantially shifts the saturation distribution toward the lower end of the spectrum, motivating the 70% saturation weighting.

4 Results

4.1 Stage 1: Detection Results

Table **??** reports the final Stage 1 metrics for all six test images. Overall precision is 0.886, recall is 0.867, and the F1-score is 0.876. The pipeline finds all sixteen balls in Table 1

and all fifteen ground-truth balls in Table 3.

Table 1: Final Stage 1 detection metrics after multi-channel LAB edge fusion.

Image	TP	FP	FN	Precision	Recall	F1-score
Table 1	16	1	0	0.941	1.000	0.970
Table 2	15	3	1	0.833	0.938	0.882
Table 3	15	0	1	1.000	0.938	0.968
Table 4	12	2	3	0.857	0.800	0.828
Table 5	10	2	4	0.833	0.714	0.769
Table 6	10	2	3	0.833	0.769	0.800
Overall	78	10	12	0.886	0.867	0.876

4.2 Stage 2: Recognition Results

Table **??** reports recognition accuracy for the true-positive detections. Across all six tables, 55 of 78 correctly localised balls receive the correct label, giving an overall recognition accuracy of 70.5%. Figure **??** shows the final labelled output for Table 4.

Table 2: Stage 2 recognition accuracy on correctly localised balls.

Image	Correct labels	True positives	Accuracy
Table 1	11	16	0.688
Table 2	9	15	0.600
Table 3	11	15	0.733
Table 4	10	12	0.833
Table 5	7	10	0.700
Table 6	7	10	0.700
Overall	55	78	0.705

4.3 Discussion

The multi-channel LAB edge-fusion detector clearly outperformed the original single-channel approach and the template-matching alternatives. The lightness channel supplies the strong outer silhouette of each ball, while the a and b channels contribute the colour boundaries that remain visible when a ball is dark, shadowed, or similar in brightness to the felt. The pocket and cushion constraints remove geometrically impossible responses, and the minimum centre separation removes duplicate detections. The resulting Stage 1 F1-score of 0.876 exceeds the previous single-channel baseline of approximately 0.831, and recall rises to 0.867.

Recognition is limited mainly by the ambiguity between similarly coloured solids and stripes. The saturation-dominant histogram improves on the K-Means attempt, but residual errors on Tables 1, 2, 5, and 6 show that a global hue and saturation distribution

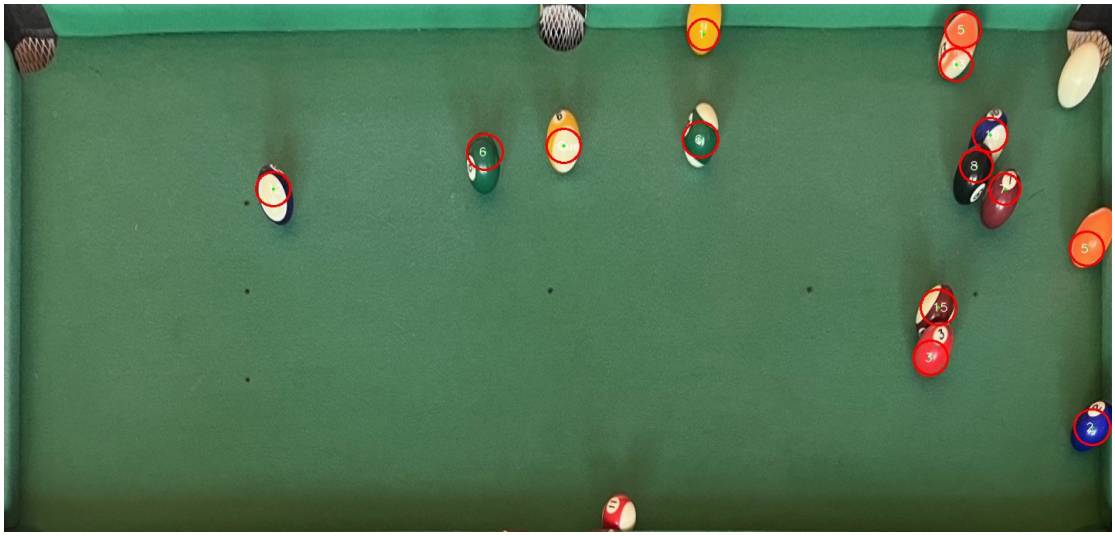


Figure 5: Final labelled output for Table 4 after Stage 2 saturation-dominant classification.

still cannot fully capture the spatial arrangement of the stripe and number. Higher-resolution descriptors that preserve spatial structure, or a dedicated stripe detector, would be the natural next step.

5 Conclusion

I replaced the failing HSV thresholding design with a multi-channel CIELAB edge-fusion detector, and I replaced the unstable K-Means recognition with a 30/70 hue–saturation histogram descriptor. The final system achieves Stage 1 precision 0.886, recall 0.867, and F1 0.876, together with Stage 2 recognition accuracy 0.705. The experiment confirms that colour-boundary fusion is more robust than static hue thresholding for this dataset, and that saturation is a more reliable cue than hue for distinguishing striped from solid balls.